



MACi ECG machine

"I have deliberately kept economics at the bottom of the list. Call it 'inversion thinking.' Economics drives everything in today's world, whereas it is merely a function of supply and demand. In my view, the concept of low cost itself is deeply flawed. What we should really concern ourselves with is lower price and, however counterintuitive it may seem, lower price is not simply lower cost. I'm not an economist, but my business sense tells me that I can drive the price down by increasing volumes and reducing profits. Besides, adopting newer business models such as crowdsourcing and collaborative manufacturing can help address this key issue. So, we must all remember that low cost is different from cost-effective, which is different from value-added. Unless the envisaged product enhances value, low cost has no meaning," he says.

From the technology standpoint, Srinivas feels the same. He says, "A low-cost product is a misnomer in my opinion. It is a product for the low-cost segment of the market but it need not be a low-quality product. Products in these segments need to be designed ground-up for success. There is no way an existing product replaced with lower-cost components can be a success here."

That said, he points out that electronic component costs have also gone down significantly in the recent past. "The mass-volume products drive down the cost of these components. Interestingly, OEMs in other industries now tend to use products or components from the PC and mobile industry to drive costs down. Processor costs for a single-core Cortex A8 at 1 GHz have gone down to as low as \$6 this

year, while two years back these commanded a healthy premium of \$20. Memory costs have also been driven down, though they fluctuate due to the changes in supply and demand. The other major component for which costs went down significantly is the display. However, for applications that need strict qualification, like in the military and automotive domain, the price erosion has been far lower."

Dorai Thodla, founder of iMorph Inc., gives us his take: "There are two ways to build a product for this market—innovation in low-cost methods, or high volume of sales with normal methods." So, you need to consider both schemes to balance value and cost for this market.

How about something like...

If you wish to get into making or selling tech to the masses, look to these interesting examples suggested by our experts for motivation.

Inspiring DIY stories. "There is something called the Maker Movement in the US. It is a self-help culture of building things you need. People who are involved or touched by the Maker Movement build simple prototypes and turn them into low-volume products (initially). Many of them involve electronics. Of late, I have been hearing of many Indian examples too," remarks Thodla. For example, a student created an app that couples a moisture sensor with an Arduino, which tweets when the moisture content is below a certain level (he wanted to connect this to a drip irrigation system). A lot of innovative products are being powered by Arduino (and its variants), Raspberry Pi and others. Plus, a variety of sensors attached to mobile devices are serving as low-cost diagnostic and healthcare tools. Solar lamps, cookers and the like are also becoming popular themes, for both start-ups and independent innovators. "Though expensive initially, these will eventually lower running costs (due to the reduction in oil and electricity usage) and make sense for the BoP. The next generation of these devices will be

enabled by intelligence (such as lamps that turn on/off based on ambient light)," says Thodla.

MACi ECG machine. "GE India has been a pioneer in reverse innovation, leading product design for emerging markets. Its innovations are not just a cost reduction of existing designs, but completely redesigned from the ground-up to make a product that is several times cheaper and yet offers reasonable value," says Srinivas. GE's low-cost MACi ECG machine, launched a few years ago, was developed by researchers in Bengaluru after understanding the need for a low-cost, durable, portable ECG machine for low-income communities. MACi is battery-operated, portable and capable of printing in three channels. With one battery charge, it can complete 500 three-channel ECGs or 250 single-channel ECGs. So, with one battery charge, it can be used in a rural hospital without electricity for close to a month. MACi is priced at around ₹ 25,000 and can be bought through easy monthly instalments. After successful early adoption, the machine was also adapted for sale to value-oriented customers in developed nations like the US.

Lullaby Baby Warmer. Another of GE's low-cost, high-value innovations, Lullaby Baby Warmer provides 69 per cent warm-up, enabling nurses to handle new-born arrivals with much less stress. The product offers value for money and ease of use, both of which are essential for mass usage. GE has incorporated technology in the form of the Calrod heater for uniform heat distribution, which makes it easy even for novices to place the baby in the warmer. The warmer also has colour-coded safety alarms, a visually-coded control panel and well-positioned lights for easy usage. According to the company's reports, the instrument offers cost savings of up to ₹ 107,000. It consumes 60 per cent less power on start-up and 20 per cent less power over 24 hours. There are more cost savings due to Calrod's lifetime warranty, lower wattage and reliable probes. "GE's MACi ECG